Soft constraint | NRMSE summary (util.py: RMS normalization)

Median of per-IC NRMSE on the same 9 complete test ICs for every model

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	**baseline**	0.092	0.476	1.07	1.02	0.329	3.25	0.363	0.967	1.01	0	18/18
1 ANN10	**soft10k**	0.323	0.812	8.39	8.07	0.659	7.79	0.329	121	397	0	14/18
2 P-only DeepONet10	**baseline**	0.0551	0.236	1	1	0.154	1	0.999	0.994	1	0	18/18
2 P-only DeepONet10	**soft10k**	0.545	0.876	1	1	0.958	1	1	451	872	0	18/18
3 ANN10 + P-only DeepONet10	**baseline**	0.03	0.233	1	0.913	0.0322	0.998	0.317	0.992	0.999	0	18/18
3 ANN10 + P-only DeepONet10	**soft10k**	0.301	0.823	4.32	14.1	0.687	7.59	0.601	700	1.22e+04	0	18/18
4 T/P DeepONet10	**baseline**	0.0466	0.212	1.4	0.786	0.196	0.986	0.288	0.993	1	0	17/18
4 T/P DeepONet10	**soft10k**	0.6	0.909	9.45	1.06	0.943	2.59	0.971	59.3	1	0	15/18
5 ANN10 + T/P DeepONet10	**baseline**	0.0444	0.13	1.01	0.965	0.13	0.999	0.606	0.921	0.996	0	18/18
5 ANN10 + T/P DeepONet10	**soft10k**	0.342	0.968	6.14	7.32	0.655	3.78	0.656	118	372	0	18/18
6 T/P/ϕ DeepONet10	**baseline**	0.149	0.389	1	1.04	0.557	1.05	1	1	1	0	17/18
6 T/P/ϕ DeepONet10	**soft10k**	0.505	0.857	11.4	1.12	0.873	1.44	0.728	6.21	1.61	0	17/18
7 ANN10 + T/P/ϕ DeepONet10	**baseline**	0.219	0.369	1	1	0.472	1	1	0.998	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.199	0.937	4.81	5.75	0.621	2.14	0.353	74.1	375	0	18/18

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
All models have10000 updates. Baseline: mu0. Soft: mu10 element-L1. Each mixed team uses both corresponding ANN and DeepONet models.
Common ICs: 5, 7, 39, 44, 53, 64, 68, 69, 73.